

# MATH 252 – ELEMENTARY DIFFERENTIAL EQUATIONS

Rutgers University, Summer 2022

---

|                    |                                                                      |                |                    |
|--------------------|----------------------------------------------------------------------|----------------|--------------------|
| <b>Instructor:</b> | Bernardo Rivas                                                       | <b>Time:</b>   | MWF 6pm – 8:45pm   |
| <b>Email:</b>      | <a href="mailto:bad162@math.rutgers.edu">bad162@math.rutgers.edu</a> | <b>Format:</b> | Online Synchronous |

---

**Course Page:** <https://rutgers.instructure.com/courses/181780>

**Office Hours:** TBD

**Textbook:** Blanchard, Devaney, Hall, *Differential Equations*, 4<sup>th</sup> ed, Cengage, 2012.

## Learning goals:

- Analyze and model processes from physics, engineering, biology, chemistry, economy, etc. using differential equations;
- Classify and manipulate differential equations;
- Employ a combination of analytical, numerical and qualitative methods to solve differential equations.

## Grading Breakdown

- Final (30%), Midterm (20%), Quizzes (20%), Homeworks (10%), MATLAB (10%), Others (10%).

## Important Dates

|                    |                   |
|--------------------|-------------------|
| Homework due ..... | Tuesdays, 11:59pm |
| MATLAB due .....   | Fridays, 11:59pm  |
| Midterm .....      | June 15           |
| Final Exam .....   | July 8            |

**Midterm and Final:** Exams of 80 minutes and 3 hours respectively, proctored online using BigBlueButton and/or additional tools. If for any reason a student will not be able to take an exam, or finds themselves in a situation, medical or otherwise, in which they will not be able to perform at their usual proficiency, they should notify the instructor right away and explain the situation. The instructor must be notified as soon as possible, and especially before the exam. In that case, a student who is excused from the midterm exam not take a make-up. Instead, the final exam will be worth 50% of the total grade. Every student must take a final exam, and in case of excused absence a make-up final must be arranged.

**Quizzes:** A proctored online test using BigBlueButton and/or additional tools. A quiz will be given in every lecture. There is no make-up for quizzes, but the two lowest scores will be dropped.

**Homeworks:** Selection of exercises (with or without reading material) assigned to be submitted in a single pdf file on Canvas. You are allowed and encouraged to discuss with others, consult class materials and other sources (and give them proper credit), but the work you submit must be your own (do not copy).

**MATLAB assignments:** Selection of exercises assigned to be submitted in a single xml file on Canvas. Assignments are written in a way to not penalize students that are not familiar with the software. Example code will be provided and students are encouraged to ask me for help if needed. It can be obtained free of charge at <https://software.rutgers.edu/tags/tag/?tag=18>.

Table 1: **Tentative outline of the course**

| Date        | Sections           | Main Topics                                                                                                   | Exam/Quiz  |
|-------------|--------------------|---------------------------------------------------------------------------------------------------------------|------------|
| Wed, Jun 01 | 1.1, 1.2           | Modeling, Separation of Variables                                                                             | Quiz 1     |
| Fri, Jun 03 | 1.3, 1.4, 1.5      | Slope Fields, Euler's Method, Existence and Uniqueness of Solutions                                           | Quiz 2     |
| Mon, Jun 06 | 1.6, 1.7, 1.8      | Equilibria and the Phase Line, Bifurcations, Linear Equations                                                 | Quiz 3     |
| Wed, Jun 08 | 1.9, 2.3, 3.6      | Integrating Factors, Undetermined Coefficients, 2nd Order Constant Coefficient Equations, Harmonic Oscillator | Quiz 4     |
| Fri, Jun 10 | 2.1, 2.2, 2.4      | Systems: Modeling, Geometry and Analytic Methods                                                              | Quiz 5     |
| Mon, Jun 13 | 2.5, 2.6, 2.7      | Existence and Uniqueness for Systems, SIR Model, Phase Plane                                                  | Quiz 6     |
| Wed Jun 15  | N/A                |                                                                                                               | Midterm    |
| Fri, Jun 17 | N/A                |                                                                                                               |            |
| Mon, Jun 20 | 3.1, 3.2           | Linear Systems and Linearity Principle, Straight-line solutions                                               | Quiz 7     |
| Wed, Jun 22 | 3.3, 3.4           | Linear Systems: Real and Complex eigenvalues                                                                  | Quiz 8     |
| Fri, Jun 24 | 3.5, 3.7           | Linear Systems: Repeated eigenvalues, 2nd Order Equations as Systems, Trace-Determinant Plane                 | Quiz 9     |
| Mon, Jun 27 | 4.1, 4.2, 4.3, 4.4 | Forced Harmonic Oscillator                                                                                    | Quiz 10    |
| Wed, Jun 29 | 5.1, 5.2           | Linearization, Qualitative Analysis                                                                           | Quiz 11    |
| Fri, Jul 01 | 5.3, 5.4           | Hamiltonian Systems, Dissipative Systems                                                                      | Quiz 12    |
| Mon, Jul 04 | N/A                |                                                                                                               |            |
| Wed, Jul 06 | TBD                | TBD                                                                                                           | TBD        |
| Fri, Jul 08 | N/A                |                                                                                                               | Final Exam |

**Others:** Mini-quizzes, reading quizzes, poll questions, or similar format that evaluates participation in class or reading assignments.

**Department Technology Policy:** By signing up for a course in the math department, a student consents to have videos of themselves recorded as needed for authentication purposes. We take seriously our obligation to students to provide a level playing field that does not disadvantage our many students of high integrity. Therefore, in all courses, there will be some form of video control of test environments. This type of online proctoring will require the use of a webcam and a mic. Students will also need, at those times, to be in an environment that affords them privacy, if only to avoid the recording of others who may not be aware of it. <https://www.math.rutgers.edu/academics/undergraduate/1685>

**Late Submission Policy:** Unless otherwise noted, all assignments are due at the time and date listed. If you experience an unavoidable personal situation that prevents you from completing work on time, please inform me prior to the date the work is due. Late work will result in points taken off, a lowering of the assignment grade, and/or a zero grade, depending on the assignment.

**Attendance and Participation Policy:** Students are expected to attend all classes and be on time; if you expect to miss one or two classes or a period of time in an online class, please use the University absence reporting website – <https://sims.rutgers.edu/ssra/> - to indicate the date and reason for your absence. An email will automatically be sent to me from this system. Note that if you must miss classes for longer than one week, you should contact a dean of students to help verify your circumstances. **For the purpose**

of online classes, you will be considered absent from class if you do not engage on the course activities in timely fashion. I've been playing Elden Ring.

**Academic Integrity:** All students in the course are expected to be familiar with and abide by the academic integrity policy (<http://academicintegrity.rutgers.edu/academic-integrity-at-rutgers>). Violations of the policy are taken very seriously.

**Students with Disabilities:** Full disability policies and procedures are indicated at <http://ods.rutgers.edu/>. Students with disabilities requesting accommodations must present a Letter of Accommodations to the instructor as early in the term as possible (see <https://ods.rutgers.edu/my-accommodations/letter-of-accommodations>).

**Student-Wellness Services:** These services may be provided remotely or in-person during the summer. Contact the places individually for more information.

- **Counseling, ADAP & Psychiatric Services (CAPS):**
  - (848) 932-7884 – 17 Senior Street, New Brunswick, NJ 08901
  - <http://health.rutgers.edu/medical-counseling-services/counseling/>
  - CAPS is a University mental health support service that includes counseling, alcohol and other drug assistance, and psychiatric services staffed by a team of professionals within Rutgers Health services to support students' efforts to succeed at Rutgers University. CAPS offers a variety of services that include: individual therapy, group therapy and workshops, crisis intervention, referral to specialists in the community, and consultation and collaboration with campus partners.
- **Crisis Intervention:** <http://health.rutgers.edu/medical-counseling-services/counseling/crisis-intervention/>
- **Report a Concern:** <http://health.rutgers.edu/do-something-to-help/>
- **Violence Prevention & Victim Assistance (VPVA):**
  - (848) 932-1181 – 3 Bartlett Street, New Brunswick, NJ 08901
  - [www.vpva.rutgers.edu/](http://www.vpva.rutgers.edu/)
  - The Office for Violence Prevention and Victim Assistance provides confidential crisis intervention, counseling and advocacy for victims of sexual and relationship violence and stalking to students, staff and faculty. To reach staff during office hours when the university is open or to reach an advocate after hours, call 848-932-1181.